

Electromagnetism. Lecture 8

Topics:

- Introduction
- Oblique incident waves
- Definitions
- Polarization, angle of incidence, beta vector
- TLA in orthogonal directions
- Snell's law
- Intrinsic impedance in normal direction
- Brewster angle, critical angle
- Evanescent waves
- Rayleigh reflection coefficient

Reading:

GMF pages 66-104, 114, 129-153

Exercise 18.1

Calculate for the following substances:

Diesel oil, $\epsilon_r = 2$

Barium titanate stannate, $\epsilon_r = 20\,000$ (9BaTiO₃ : 1BaSnO₃)

- The critical angle from matter → air
- Brewster angle substance → air
- The Brewster angle air → matter

Exercise 18.2

A wave with a frequency of 300 MHz and a field strength of 1 V/m strikes a water surface with horizontal polarization and an angle of incidence of 60°. We have $\epsilon_r = 81$ for the water.

- Determine the angle of incidence and the angle of refraction.
- The sketch situation.
- Calculate the Rayleigh reflection coefficient.
- Calculate the strength and direction of the reflected and refracted E-field.
- Calculate strength and direction of reflected and refracted H-field.
- Calculate the wavelength in the tangential and normal direction in the water.

Exercise 18.3

Show that there are no reflections from the back of a plane-parallel pane that is hit by a light beam at the Brewster angle on the front.

Exercise 18.4

A military missile comes flying towards a city. The townspeople want to shoot the missile down with a large laser cannon, and it is therefore hit by a laser beam with a power density of 10

MW/m² in a spot with an area of 10 cm². The wavelength is 500 nm. (The idea is, of course, that the electromagnetic effect should cause the material to break down).

To annoy the Americans, however, the missile is silver-plated. Assuming that the silver layer is thick, and that the laser beam falls perpendicularly into the surface, calculate the reflected power and the power absorbed by the missile.

$$\epsilon_r = \mu_r = 1, \sigma = 61,2 \times 10^6 \text{ S/m.}$$

Exercise 18.5

a. U.S. Standard for military personnel in microwave environment (previously also used by it Danish Defense) says that one must be exposed to a radiation of no more than 10 mW/cm². Calculate the corresponding E and H field.

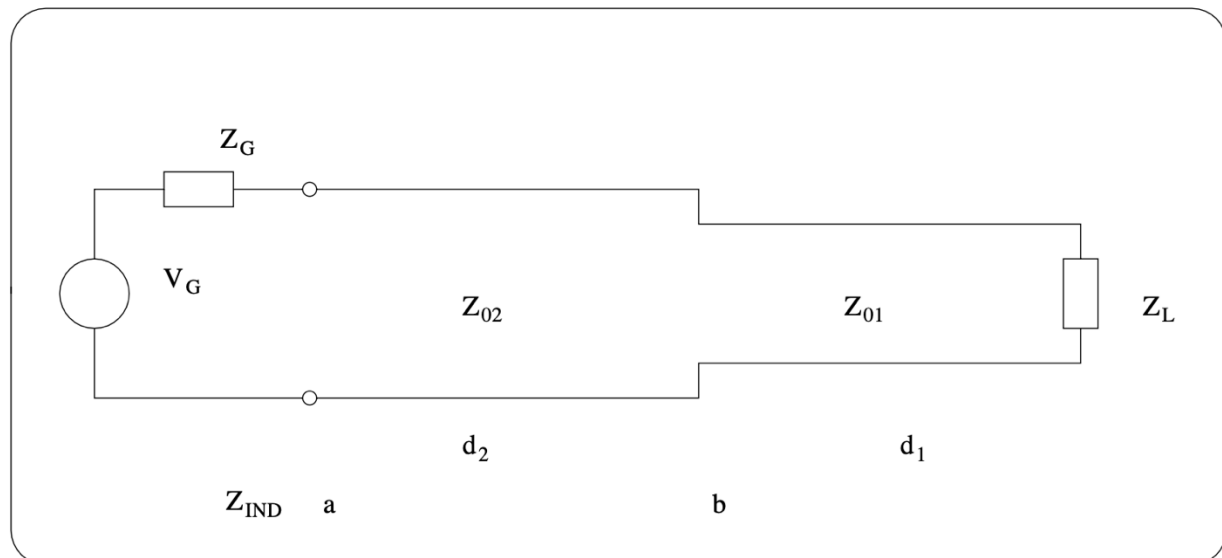
b. On a sunny day, the earth receives about 1,3kW/m² of broad-spectrum radiation from the Sun. If we assume that this radiation arrives as a plane wave at a single frequency, what will be the corresponding E and H field?

Exercise 18.6

For the setup below we have:

$$Z_{01} = 75 \Omega, Z_{02} = 240 \Omega, Z_L = (8 - j38) \Omega, d_1 = 0,7\lambda, d_2 = 0,8\lambda$$

Both cables are considered lossless.



- Determine the value of Z_{IN}
- When $V^+ = 10 \text{ V}$ (eff.) immediately to the right of a , then calculate V^+ and V^- immediately to the right of b
- Calculate the incident and the reflected power immediately to the right of a
- Calculate the incident and the reflected power immediately to the right of b
- How much total power is delivered from the generator into the transmission line?
- How much power is allocated in Z_L ?

g. Show that we get $V^+ = 10 \text{ V (eff.)}$ immediately to the right of b , if we have $Z_G = 240 \Omega$ and $V_G = 20 \text{ V (eff.)}$.

One can conveniently use the following notation:

$K(a^+) =$ the reflection coefficient just to the right of a

$K(a^-) =$ the reflection coefficient just to the left of a

Solutions:

Exercise 18.1	45°	0,4°	35,3°	0,4°	54,7°	89,6°	
Exercise 18.2	60°	5,52°	In the yz-plane	-0,8943	0,894 V/m $\hat{z}$	-0,1057 V/m $\hat{z}$	2,372
	mA/m 60° relative to the interface		2,524 mA/m	5,5° relative to the interface	1,16m	11,16cm	
Exercise 18.3	There are not.						
Exercise 18.4	9,361W 639W						
Exercise 18.5	194,2 V/m eff.		0,515 A/m eff.	700,1 V/m eff.	1,857 A/m eff.		
Exercise 18.6	34,6 -j338,9 Ω		4,34 < 96° V		3,67 < 174° V		